

Chapter 3: The Transport Layer

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EXERCISE 1

Consider transferring an enormous file of L bytes from Host A to Host B. Assume an MSS of 536 B.

a) What is the maximum value of L such that TCP sequence numbers are not exhausted? Recall that the TCP sequence number field has (4B) Total sequence number field = 2^{32} (4B)

b) For the L you obtain in (a), find how long it takes to transmit the file. Assume that a total of 66 bytes of transport, network, and data-link header are added to each segment before the resulting packet is sent over a 155 Mbps link. Maximum segment size

$L_{\max} = 2^{32} \text{ byte} \approx 4.19 \text{ GB}$
 • Each TCP segment: $536 \text{ Bytes} + 66 = 602 \text{ bytes}$
 • We got $L_{\max} = 2^{32} \text{ bytes}$ from (a): # Packets = $\frac{2^{32} \text{ bytes}}{602 \text{ bytes}} \approx 8016731 \text{ Pkt}$
 • Total data Transmitted over link = $8,016,731 \times 602 \approx 4.83 \text{ GB}$

EXERCISE 2

Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.

a) In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?

b) If the first segment arrives before the second segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number, the source port number, and the destination port number?

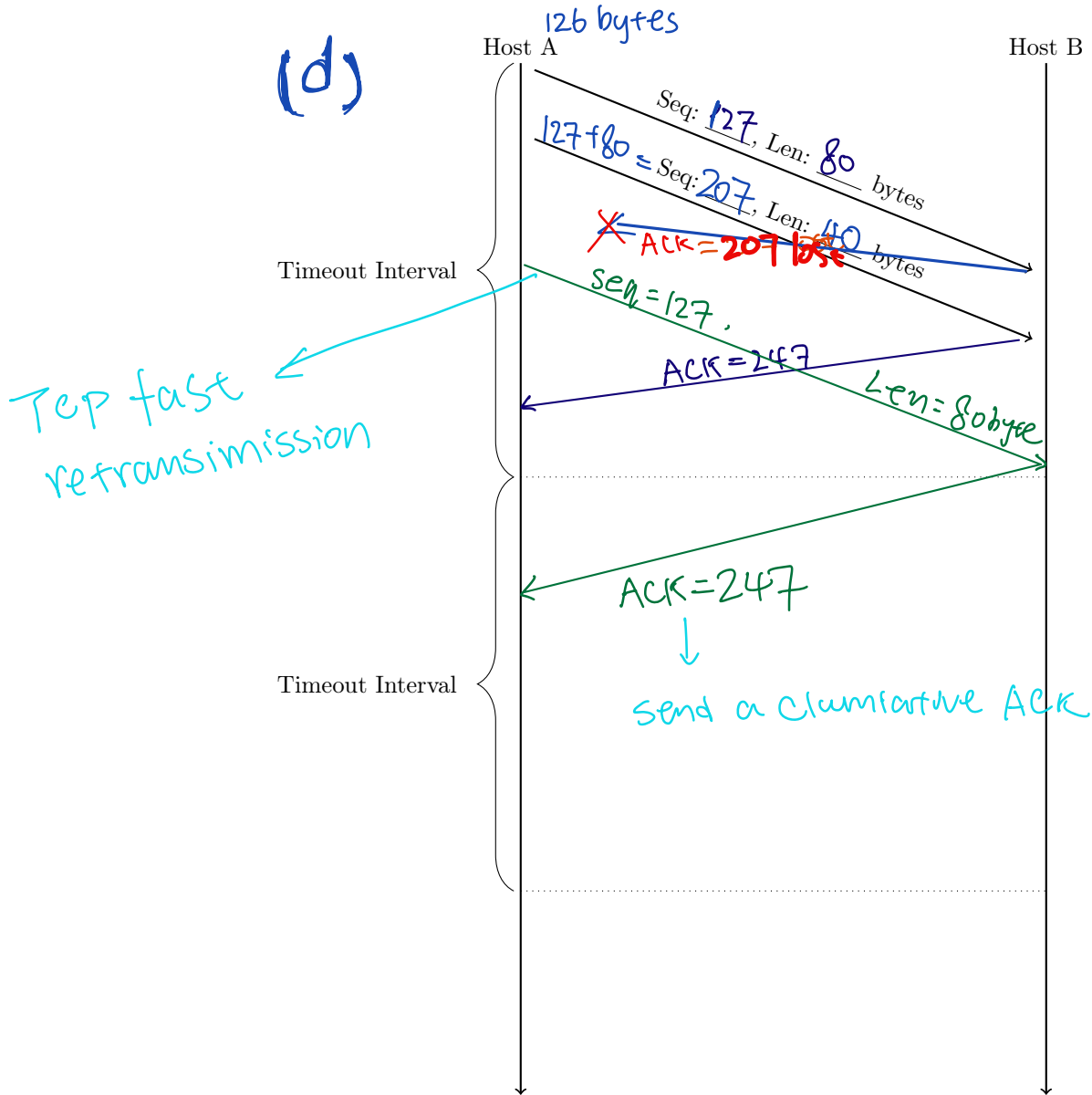
c) If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number?

d) Suppose the two segments sent by A arrive in order at B. The first acknowledgment is lost and the second acknowledgment arrives after the first timeout interval. Draw a timing diagram, showing these segments and all other segments and acknowledgments sent. (Assume there is no additional packet loss.) For each segment in your figure, provide the sequence number and the number of bytes of data; for each acknowledgment that you add, provide

(a) sequence number: $127 + 80 = 207$, source port: 302, dest port: 80

(b) ACK number: 207 (first sequence #), source port: 302, dest port: 80

(c) ACK # = 127 (the second segment starts 207 & Len = 40 byte but since byte 127 - 206 is missing, Host B will acknowledge the first missing byte. 127)



the acknowledgment number.

EXERCISE 3

Consider the TCP procedure for estimating RTT. Let $SampleRTT_1$ be the most recent sample RTT, let $SampleRTT_2$ be the next most recent sample RTT, and so on. Suppose that $\alpha = 0.1$.

- a) For a given TCP connection, suppose four acknowledgments have been returned with corresponding sample RTTs: $SampleRTT_4$, $SampleRTT_3$, $SampleRTT_2$, and $SampleRTT_1$. Express $EstimatedRTT$ in terms of the four sample RTTs.
- b) Generalize your formula for n sample RTTs.
- c) For the formula in part (b), let n approach infinity. Comment on why this averaging is called an exponential moving average.

a) Given equation is: $\text{Estimated RTT} = (1-\alpha)\text{Estimated RTT} + \alpha * \text{Sample RTT}$
since $\alpha = 0.1$, thus

$$\text{Estimated RTT} = 0.9 * \text{Estimated RTT} + 0.1 * \text{Sample RTT}$$

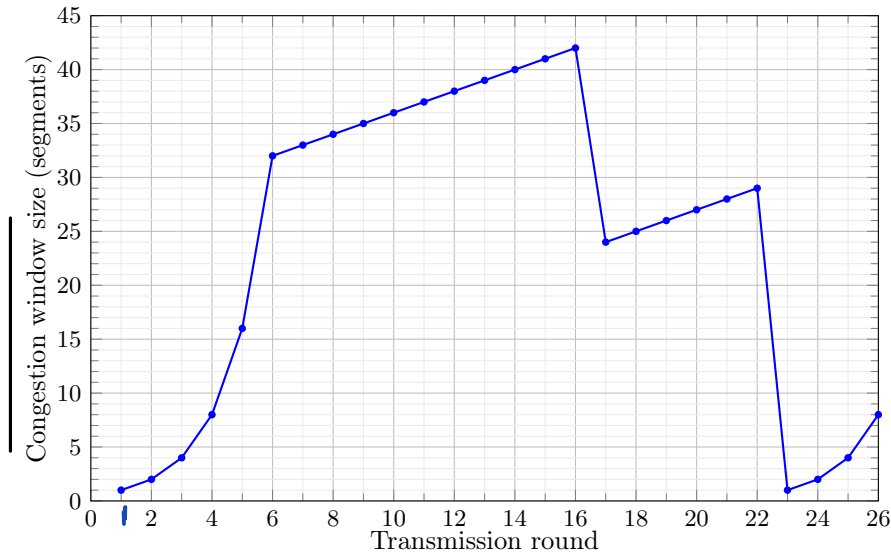
Therefore, Estimated RTT in terms of the four sample RTTs:

$$\text{Estimated RTT} = 0.9^3 * \text{Sample RTT}_4 + 0.9^2 * \text{Sample RTT}_3 + 0.9^1 * \text{Sample RTT}_2 + 0.9^0 * \text{Sample RTT}_1$$

$$\text{Estimated RTT}_n = 0.9^{n-1} * \text{Sample RTT}_n + 0.9^{n-2} * \text{Sample RTT}_{(n-1)} + \dots + 0.9^1 * \text{Sample RTT}_2 + 0.9^0 * \text{Sample RTT}_1$$

c) For the formula in part (b), when $n \rightarrow \infty$, the term $0.9^{n-1}, 0.9^{n-2}, 0.9^{n-3} \dots 0.9^1$ becomes smaller and smaller, which means that the older Sample RTTs have less influence on the Estimated RTT. The averaging procedure is called an exponential moving average because it gives more weight to recent sample RTTs, resulting a smoothed and updated estimation of the RTT.

EXERCISE 4



Consider the figure above. Assuming TCP Reno is the protocol experiencing the behavior shown, answer the following questions. In all cases, you should provide a short discussion justifying your answer.

- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of *ssthresh* at the first transmission round?
- What is the value of *ssthresh* at the 18th transmission round?
- What is the value of *ssthresh* at the 24th transmission round?
- During what transmission round is the 70th segment sent?
- Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of *ssthresh*?
- Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the *ssthresh* and the congestion window size at the 19th round?
- Again suppose TCP Tahoe is used, and there is a timeout event at 22nd round. How many packets have been sent out from 17th round till 22nd round, inclusive?

a) TCP slow start grows exponentially. thus from the graph above TCP slow start operates in interval $[1, 6]$ and $[23, 26]$

b) The intervals time from 6 to 16 and from 17 to 22 is when TCP congestion avoidance is operating. because congestion avoidance grows linearly

c) After 16th transmission round, the congestion window drops suddenly after 16th round, but not to 1, this means TCP reno detected packet loss using triple duplicate ACKs, which trigger fast recovery instead of time out.

d) After 22th transmission round, the congestion window drops to 1, which means the loss was detected by a timeout, causing TCP reno to start slow start.

e) The initial value of ssthresh at the first transmission round is 32.

f) The value of ssthresh at the 16th transmission round is 42, $ssthresh = cwnd/2 = 42/2 = \underline{21}$

g) When TCP Reno detects loss by a timeout, it sets $ssthresh = cwnd/2$ at it also resets cwnd to 1. When transmission round is 22, the congestion window size is 24. Hence the $ssthresh = \underline{14}$ during the 24th transmission round.

h) after 6 round : $1 + 2 + 4 + 8 + 16 + 32 = 63$ segments

Round : $cwnd = 33 \rightarrow$ sends 33 segments

The remaining $70 - 63 = 7$ segments are sent in the next round. Thus, the packet TO is sent during the 7th transmission round

i) The threshold will be set to half the current value of the congestion window (8) when the loss occurred and congestion window will be set to the new threshold + 3MSS. Therefore the new value of the threshold should be 4 and window size should be 7

j) $ssthresh = 21$, congestion window size is 1

k) Since a timeout at round 22 resets cwnd to 1, we count the number of packets sent from round: 17 to 22.

$$1 + 2 + 4 + 8 + 16 + 21 = \underline{52}$$

thus, 52 Packets have been sent from rounds 17 to 22.